

MEDIATEK

MT8668 Yocto USB User Manual

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1 USB

1.1 概述

1.1.1 基本概述

本章节介绍主要 MT8668 USB 模块信息，以及 USB 常见问题以及分析方法。

1.1.2 缩略词

表 1-1. 缩略词

缩略词	全称	释义
ADB	Android Debug Bridge	是一个多功能命令行工具，允许用户与 Android 设备进行通信
MTU3	MediaTek USB3.0	SSUSB IP 中的 device controller
PHY	Physical Layer	USB 的物理层，负责实际的电气和物理信号传输
SSUSB	Super Speed USB	MediaTek 设计的 USB IP，支持 USB3.0 的 host/device 双角色控制器
USB	Universal Serial Bus	通用串行总线
xHCI	eXtensible Host Controller Interface	SSUSB IP 中的 host controller

1.2 USB 硬件架构

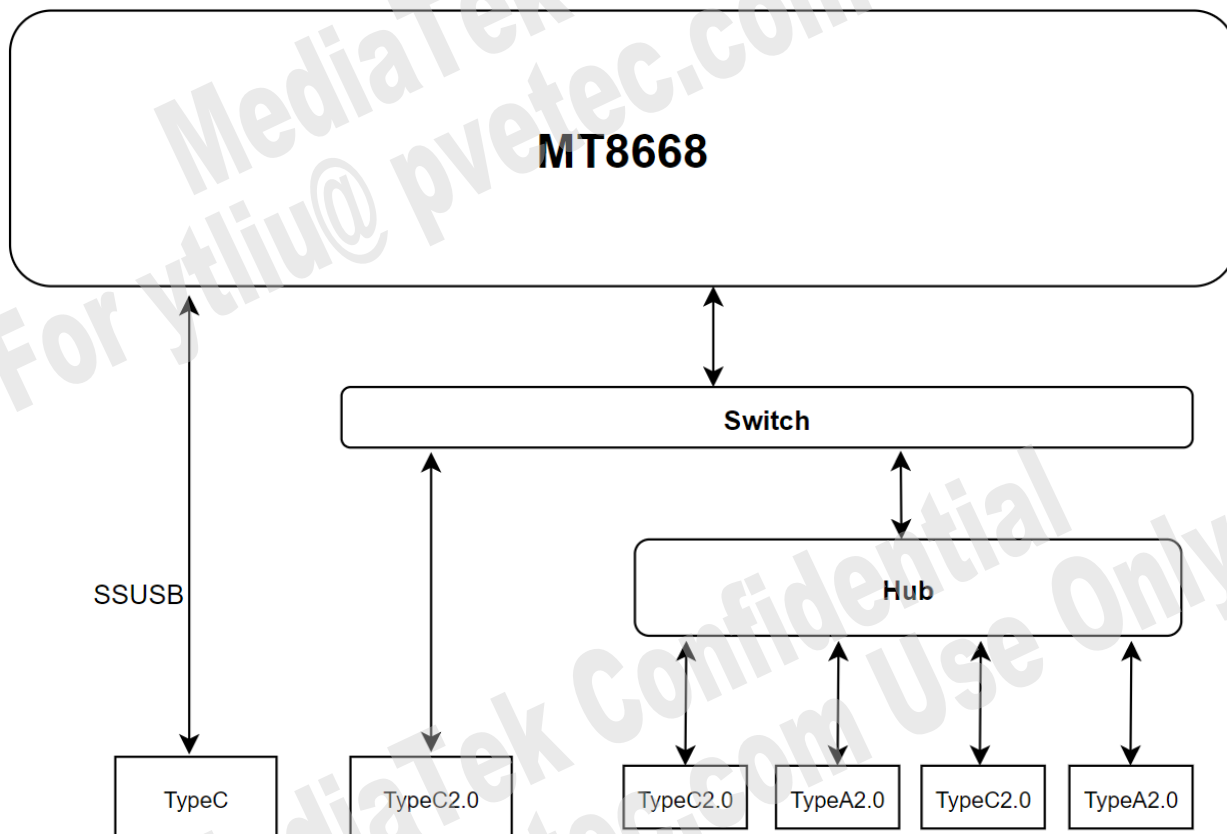


图 1-1. USB 架构

- Port0: 最高支持 USB3.2 gen1, 5Gbps, host/device dual role controller
- Switch: 最高支持 USB2.0 480Mbps, 能实现 Hub 与 USB2.0 之间的切换
- Hub: 最高支持 4 个 USB 口

1.3 配置/客制化指南

1.3.1 DTS 配置

1.3.1.1 SSUSB 节点信息

```
ssusb: usb0@11201000 {
    compatible = "mediatek,mtu3";
    reg = <0 0x11201000 0 0x2e00>,
        <0 0x11203e00 0 0x0100>;
    reg-names = "mac", "ippc";
    vusb33-supply = <&mt6368_vusb>;
    interrupts-extended = <&gic GIC_SPI 512 IRQ_TYPE_LEVEL_HIGH 0>,
        <&pio 201 IRQ_TYPE_LEVEL_HIGH>;
    interrupt-names = "device", "wakeup";
    phy-cells = <1>;
    phys = <&u2port0 PHY_TYPE_USB2>;
    clocks = <&cksys_reg_clk CLK_CKSYS_REG_USB_TOP_SEL>,
        <&cksys_reg_clk CLK_CKSYS_REG_USB_XHCI_SEL>,
        <&cksys_reg_clk CLK_CKSYS_REG_USB_FRMCNT_SEL>;
    clock-names = "sys_ck", "host_ck", "frmcnt_ck";
    power-domains = <&scpsys MT6881_POWER_DOMAIN_SSUSB>;
    #address-cells = <2>;
    #size-cells = <2>;
    ranges;
    dr_mode = "otg";
    maximum-speed = "high-speed";
    mediatek,force-vbus;
    mediatek,clk-mgr;
    mediatek,hwrscs-vers = <3>;
    mediatek,noise-still-tr;
    mediatek,syscon-wakeup = <&pericfg_ao_reg_clk 0x214 108>;
    wakeup-source;
    usb-role-switch;
    cdp-block;
    usb-psy-name = "mtk-master-charger";
    typec-name = "rt-pd-manager";
    typec-port-name = "port0";

    port {
        mtu3_drd_switch: endpoint {
            remote-endpoint = <&usb_role>;
        };
    };

    usb_host: xhci0@11200000 {
        compatible = "mediatek,mtk-xhci", "mediatek,mt6881-xhci";
        reg = <0 0x11200000 0 0x1000>;
        reg-names = "mac";
        interrupts = <GIC_SPI 513 IRQ_TYPE_LEVEL_HIGH 0>;
        clocks = <&clk26m>;
        clock-names = "sys_ck";
        status = "okay";
    };
};
```

```
&ssusb {
    pinctrl-names = "default";
    pinctrl-0 = <&ssusb port0_default>;
    vusb33-supply = <&mt6373_vusb>;
    role-switch-default-mode = "peripheral";
};
```

图 1-2. mt6881.dts 和 auto8668p1_64.dts 中的 ssusb 节点

ssusb & usb_host DTS node 中各个 property 字段含义：

- **ssusb**: USB device controller MTU3 节点
- **phys**: USB 使用的 u2/u3 phy
- **dr_mode**: USB dual role mode, 可以设置为 otg/peripheral/host
- **maximum-speed**: device controller 使用的速度, 可以设置为 super-speed/high-speed/full-speed
- **usb-role-switch**: USB role 切换采用 Linux 标准的 usb-role-switch get/set ops
- **role-switch-default-mode**: 配合 usb-role-switch 设置 IP default USB mode, 可以设置为 otg/peripheral/host
- **mediatek,clk-mgr**: 切换到 USB none/device 时, host driver 会卸载以达到更加省电的目的, 否则 host driver 会保留
- **mediatek,force-vbus**: 强制 device controller 认为 VBUS 存在, 用于未连接 vbusvalid 引脚的 PCB。
- **usb_host**: USB host controller xHCI 节点

1.3.1.2 USB PHY 节点

```

u2phy: usb-phy0@11f40000 {
    compatible = "mediatek,xsphy", "mediatek,mt6881-xsphy";
    reg = <0 0x11f40000 0 0x400>;
    reg-names = "u2_port_base";
    #address-cells = <2>;
    #size-cells = <2>;
    ranges;

    u2port0: usb2-phy0@11f40000 {
        reg = <0 0x11f40000 0 0x400>;
        clocks = <&clk26m>;
        clock-names = "ref";
        #phy-cells = <1>;
        mediatek,efuse-intr = <0x23>;
        mediatek,efuse-term = <0x8>;
        usb2uart;
        nvmem-cells = <&u2_phy_data>, <&u2_phy_data>;
        nvmem-cell-names = "intr_cal", "term_cal";
        nvmem-cell-masks = <0x3f 0xf00>;
        mediatek,uart-usb-sel = <&pericfg_ao_reg_clk 0xb24 0x1 1>;
    };
};

```

```

u3phy: usb3-phy0@11f53000 {
    compatible = "mediatek,xsphy", "mediatek,mt6881-xsphy";
    reg = <0 0x11f53000 0 0x200>;
    tx-chirp-capable;
    #address-cells = <2>;
    #size-cells = <2>;
    ranges;
    status = "disabled";

    u3port0: usb3-phy0@11f53000 {
        reg = <0 0x11f53400 0 0x500>;
        clocks = <&clk26m>;
        clock-names = "ref";
        #phy-cells = <1>;
        mediatek,u3-rx-fix;
        mediatek,u3-gen2-hqa;
    };
};

```

图 1-3. mt6881.dts 中的 USB PHY 节点

u2phy & u3phy DTS node 中各个 property 字段含义：

- **u2phy**: USB2.0 PHY 节点
- **u3phy**: USB3.0 PHY 节点

1.3.2 Kernel 配置和 ko 相关信息

表 1-2. USB 相关 kernel config, ko name 以及对应 code path

Kernel Config	ko Name	Code Path
CONFIG_DEVICE_MODULES_USB_MTU3	mtu3.ko	co_device_module/drivers/usb/mtu3/
CONFIG_DEVICE_MODULES_USB_XHCI_MTK	xhci-mtk-hcd-v2.ko	co_device_module /drivers/misc/mediatek/usb/usb_xhci/
CONFIG_DEVICE_MODULES_PHY_MTK_XSPHY	phy-mtk-xspdy.ko	co_device_module /drivers/phy/mediatek/

1.3.3 Host/Device 模式切换流程

MT8668 软件支持手动切换和自动切换两种模式。

1.3.3.1 手动切换

手动切换是通过 usb-role-switch 创建的 role 节点, 手动写入 CMD 来切换 host/device/none:

Path: /sys/class/usb_role/11201000.usb0-role-switch/role

- 切换到 host mode: echo host > /sys/class/usb_role/11201000.usb0-role-switch/role
 - echo host > /sys/.../role => usb_role_switch_set_role(USB_ROLE_HOST) => mtu3, ssusb_role_sw_set => usb switch to host
- 切换到 device mode: echo device > /sys/class/usb_role/11201000.usb0-role-switch/role
 - echo device > /sys/.../role => usb_role_switch_set_role(USB_ROLE_DEVICE) => mtu3, ssusb_role_sw_set => usb switch to device
- 切换到 none mode: echo none > /sys/class/usb_role/11201000.usb0-role-switch/role
 - echo none > /sys/.../role => usb_role_switch_set_role(USB_ROLE_NONE) => mtu3, ssusb_role_sw_set => usb switch to none

1.4 常见问题/疑难解答

1.4.1 USB 相关日志

1.4.1.1 USB 初始化相关日志

Keyword: mtu3 & xhci & ssusb & xsphy

```
[ 3.852111][T10200000234] mtu3_init_threa: mtu3 11201000.usb0: uwk - reg:0x214,
version:108 //mtu3 probe start
[ 3.853328][T10200000234] mtu3_init_threa: mtu3 11201000.usb0: supply vbus not found,
using dummy regulator
[ 3.855731][T10200000234] mtu3_init_threa: mtu3 11201000.usb0: dr_mode: 3, drd: auto
[ 3.857031][T10200000234] mtu3_init_threa: mtu3 11201000.usb0: u2p_dis_msk: 0,
u3p_dis_msk: 0
[ 3.861439][T10200000234] mtu3_init_threa: mtu3 11201000.usb0: only one power domain.
[ 3.864093][T10200000234] mtu3_init_threa: mtu3 11201000.usb0: sw chip version:0
[ 3.866224][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0: device src:-22
vrt:-22 term:-22 rev6:-22
[ 3.868288][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0: host src:-22
vrt:-22 term:-22 rev6:-22
[ 3.874056][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0: u2_intr:35
intr_ofs:-64 host_intr_ofs:-64 discth:-22
[ 3.875476][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0: term_cal:8
term_ofs:-16 host_term_ofs:-16
[ 3.876771][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0: rx_sqth:-22
host_rx_sqth:-22
[ 3.879081][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0: rx_sqd:-22
host_rx_sqd:-22
[ 3.881332][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0: pll_fbksel:-22,
pll_posdiv: -22
[ 3.883579][T10200000234] mtu3_init_threa: mtk-xsphy 11f53000.usb3-phy0: u3_intr:-22, tx-
imp:-22, rx-imp:-22
[ 3.885902][T10200000234] mtu3_init_threa: mtk-xsphy 11f53000.usb3-phy0: tx-lctxcml:-22,
tx-lctxc0:-22, tx-lctxcpl:-22
[ 3.888170][T10200000234] mtu3_init_threa: mtu3 11201000.usb0: phy_u2_device_props=0,
phy_u2_host_props=0
[ 3.890246][T10200000234] mtu3_init_threa: mtk-xsphy 11f40000.usb-phy0:
u2_phy_instance_power_on(0)
[ 3.892124][T10200000234] mtu3_init_threa: mtk-xsphy 11f53000.usb3-phy0:
u3_phy_instance_power_on apply u3_gen2_hqa.
[ 3.893637][T10200000234] mtu3_init_threa: mtk-xsphy 11f53000.usb3-phy0:
u3_phy_instance_power_on(0) //phy power on
[ 3.895899][T10700000234] mtu3_init_threa: mtk-xsphy 11f53000.usb3-phy0: Disable
tx_chirpK.
[ 3.898249][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: wakeup irq 242
[ 3.901885][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: usb3-drd: 1
[ 3.902790][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: ssusb_hwrscs_req_v2_v3
state=1, spm_ctrl=0x2025007f
[ 3.906200][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: irq 382
```

```

[ 3.907338][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: vbus_draw_enable: 0
[ 3.909906][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: typec: rt-pd-manager
[ 3.910925][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: typec port: port0
[ 3.912943][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: IP version 0x1015(U3 IP)
[ 3.914121][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: fifosz/epnum: Tx=0x2000/8,
Rx=0x2000/8
[ 3.915308][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: disable accept_lgo
[ 3.916259][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: max_speed_host: super-
speed //max speed 支持 super-speed
[ 3.918212][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: u2_lpm_quirks: 0x0
[ 3.918217][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: dma mask: 36 bits
[ 3.922612][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: supply vbus not
found, using dummy regulator //xhci probe start
[ 3.924999][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: supply vusb33 not
found, using dummy regulator
[ 3.927741][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: xHCI Host Controller
[ 3.928826][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: new USB bus
registered, assigned bus number 1
[ 3.931508][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: hcc params 0x01440f91
hci version 0x120 quirks 0x0000000000200010
[ 3.933684][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: irq 383, io mem
0x11200000
[ 3.936072][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: xHCI Host Controller
[ 3.937102][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: new USB bus
registered, assigned bus number 2
[ 3.939187][T10700000234] mtu3_init_threa: xhci-mtk 11200000.xhci0: Host supports USB 3.2
Enhanced SuperSpeed
[ 3.940465][T10700000234] mtu3_init_threa: usb usb1: New USB device found, idVendor=1d6b,
idProduct=0002, bcdDevice= 6.12
[ 3.943929][T10700000234] mtu3_init_threa: usb usb1: New USB device strings: Mfr=3,
Product=2, SerialNumber=1
[ 3.945993][T10700000234] mtu3_init_threa: usb usb1: Product: xHCI Host Controller
[ 3.945996][T10700000234] mtu3_init_threa: usb usb1: Manufacturer: Linux 6.12.38-
android16-5-g477adbee20b6-4k xhci-hcd
[ 3.945998][T10700000234] mtu3_init_threa: usb usb1: SerialNumber: 11200000.xhci0
[ 3.946439][T10700000234] mtu3_init_threa: hub 1-0:1.0: USB hub found
[ 3.948222][T10700000234] mtu3_init_threa: hub 1-0:1.0: 1 port detected
[ 3.950917][T10700000234] mtu3_init_threa: usb usb2: We don't know the algorithms for LPM
for this host, disabling LPM.
[ 3.952617][T10700000234] mtu3_init_threa: usb usb2: New USB device found, idVendor=1d6b,
idProduct=0003, bcdDevice= 6.12
[ 3.954502][T10700000234] mtu3_init_threa: usb usb2: New USB device strings: Mfr=3,
Product=2, SerialNumber=1
[ 3.957038][T10700000234] mtu3_init_threa: usb usb2: Product: xHCI Host Controller
[ 3.959622][T10700000234] mtu3_init_threa: usb usb2: Manufacturer: Linux 6.12.38-
android16-5-g477adbee20b6-4k xhci-hcd
[ 3.962088][T10700000234] mtu3_init_threa: usb usb2: SerialNumber: 11200000.xhci0
[ 3.964636][T10700000234] mtu3_init_threa: hub 2-0:1.0: USB hub found
[ 3.966877][T10700000234] mtu3_init_threa: hub 2-0:1.0: 1 port detected
[ 3.970228][T10700000234] mtu3_init_threa: mtu3 11201000.usb0: xHCI platform device
register success... //xhci probe done

```

1.4.2 常见问题以及调试

1.4.2.1 adb 不识别调试流程

1. PC 设备管理器有认到 USB 设备, 但是 adb devices 没有设备显示

- 1) 可能是 PC 驱动有问题, 可以在设备管理器上手动更新驱动为 Android ADB Interface
- 2) 可能是 serial number 没有发送, 可以检查下面节点是否有值, 如果没有值, 说明是序列号没有写入, 可以手动 echo 0123456789ABCDEF 到这个节点, 然后重新插拔 USB 线
 - Android: `cat /config/usb_gadget/g1/strings/0x409/serialnumber`
 - Yocto: `cat /sys/kernel/config/usb_gadget/g1/strings/0x409/serialnumber`

2. PC 设备管理器没有 USB 设备连接

- 1) 确认当前 USB role 是否有切到 device mode

```
cat /sys/class/usb_role/12001000.usb0-role-switch/role
```

得到回显应该是 device, 如果是 host/none, 说明没有切换到 device mode, 可以重新插拔 usb 线, 或者手动 echo device 到这个节点来切换。

- 2) 检查 USB configs 配置是否有 ffs.adb

- Android: `ls -al /config/usb_gadget/g1/configs/b.1/`
- Yocto: `ls -al /sys/kernel/config/usb_gadget/g1/configs/c.1/`

- 3) 检查 adbd 是否正常启动

- Android: `ps -a | grep adbd`
- Yocto: `ps -e | grep adbd`

- 4) 检查 PC adb service 版本是否过旧

- `adb -version` 看下当前 adb 版本, 建议使用 1.0.39 以上
- `adb kill-server` 后再次尝试

1.4.2.2 Host 无法识别 Device

1. 确认 USB 是否有切换到 host mode

```
cat /sys/class/usb_role/12001000.usb0-role-switch/role
```

得到回显应该是 host, 如果是 device/none, 说明没有切换到 host mode, 可以重新插拔 OTG 线, 或者手动 echo host 到这个节点来切换。

2. 确认 xHCI driver 是否有成功挂载

1) 按照 1.4.1.1 小节确认是否有 xHCI probe log

2) 是否有新设备接入的 log，例如：

```
usb 1-1 new high-speed USB device number 2 using xhci-mtk ...
```

3. 确认 USB device 是否有绑定到 driver

```
cat /sys/kernel/debug/usb/devices
```

```
T: Bus=01 Lev=00 Prnt=00 Port=00 Cnt=00 Dev#= 1 Spd=480 MxCh= 1
B: Alloc= 0/800 us ( 0%), #Int= 0, #Iso= 0
D: Ver= 2.00 Cls=09(hub ) Sub=00 Prot=01 MxPS=64 #Cfgs= 1
P: Vendor=1d6b ProdID=0002 Rev= 6.06
S: Manufacturer=Linux 6.6.12 xhci-hcd
S: Product=xHCI Host Controller
S: SerialNumber=16710000.xhci1
C:* #Ifs= 1 Cfg#= 1 Atr=e0 MxPwr= 0mA
I:* If#= 0 Alt= 0 #EPs= 1 Cls=09(hub ) Sub=00 Prot=00 Driver=hub // driver = hub,
serialnumber=xhci 说明这个是我们 xhci host 的 roothub
E: Ad=81(I) Atr=03(Int.) MxPS= 4 Iv1=256ms

T: Bus=01 Lev=01 Prnt=01 Port=00 Cnt=01 Dev#= 2 Spd=480 MxCh= 0
D: Ver= 2.00 Cls=00(>ifc ) Sub=00 Prot=00 MxPS=64 #Cfgs= 1
P: Vendor=090c ProdID=1000 Rev=11.00
S: Manufacturer=INNOTECH
S: Product=GIGAStick
S: SerialNumber=AA04012700022400
C:* #Ifs= 1 Cfg#= 1 Atr=80 MxPwr=100mA
I:* If#= 0 Alt= 0 #EPs= 2 Cls=08(stor.) Sub=06 Prot=50 Driver=usb-storage // driver = usb-
storage, 说明是 U 盘设备
E: Ad=81(I) Atr=02(Bulk) MxPS= 512 Iv1=0ms
E: Ad=02(O) Atr=02(Bulk) MxPS= 512 Iv1=31875us
```

如果有遇到 Driver=none 的这种情况，说明设备正常识别，但是没有找到对应的 class driver。可能是这种设备需要开启特别的 kernel config，可以网上搜索如何开启，或者询问 MediaTek。

4. 确认其他因素

- 1) USB VBUS 是否正常开启，可以量测 port 口是否有 5V VBUS
- 2) 这个设备在其他平台，PC 上是否可以正常识别，避免是设备损坏导致不识别的问题
- 3) 信号质量不佳的设备，可以通过加 USB hub 转接的方式来测试是否可以连接

1.4.2.3 眼图测试相关

1. USB host 眼图测试节点和命令

支持的 CMD 如下：

- 1) Test packet: 测试眼图

- 2) Test J: 测试 test j
- 3) Test K: 测试 test k
- 4) Test SEO NAK: 测试 SEO NAK

2. 注意事项

- 1) 测试前, 请确保 USB 已经切换到 host mode
- 2) Test packet 只能由 roothub port 打出, 外接 hub 无法转发。请联系 hub 厂商咨询如何测试 hub downstream port 眼图
- 3) 输入 test 命令后, xHCI ip 将进入 test mode, 此时插拔 U 盘等设备是无法识别的, 无需担心, 将平台重启后, USB host 功能仍然可以使用

附件一 附加条款

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